



Project

It is required to use Matlab, or any other programming language, to perform the following task:

Echo generation and removal

In this task, you are given a speech signal and you are required to add an echo effect. Then, you will remove the echo using frequency domain techniques.

- Read the file 'audio1.wav'.
- The required echo effect consists of the original signal and 3 delayed and shifted versions:

$$y(t) = x(t) + 0.9x(t-1) + 0.8x(t-2) + 0.7x(t-3)$$

Note that the delay is in seconds in the above equation, however, you will work with the discrete-time signal $x[n]$ so you will need to determine the value of delay in samples in the equation of $y[n]$.

- Find and plot the impulse response of the echo generation system $h[n]$.
- Obtain the output $y[n]$ of the echo system using convolution. Save the file as 'echo.wav'.
- Now, we will remove the echo using frequency domain techniques.

The spectrum of a discrete-time signal (the discrete-time Fourier transform or DTFT) is generally a continuous function of frequency, which is periodic with a period of 2π . Therefore, it is not suitable for numerical computations. Instead, we will use the **Discrete Fourier Transform (DFT)**, which is a sampled version of the DTFT.

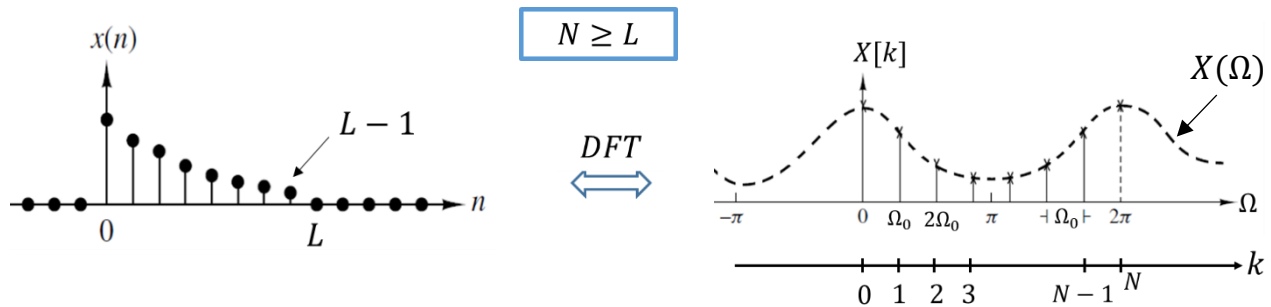


Fig. 1 DFT

DFT represents a time-domain signal of length L using N samples of its DTFT spectrum (from 0 to $N-1$) in the range of 0 to 2π , as shown in Fig. 1. Note that N must be greater than or equal to L . This is called N -point DFT. In the figure, $X(\Omega)$ is the DTFT, while $X[k]$ is the DFT.

We say that $X[k]$ is the DFT of $x[n]$ and $x[n]$ is the inverse DFT (IDFT) of $X[k]$. These two operations are implemented in Matlab using 'fft' and 'ifft' functions, respectively. Fast Fourier transform (FFT) is an efficient algorithm for computing the DFT.

Now, given $h[n]$ and $y[n]$, you are required to remove the echo effect and obtain $x[n]$.

We will perform this operation in the frequency domain. It should be noted that $y[n]$ has length $L_y = L_x + L_h - 1$, therefore, $X[k]$, $H[k]$ and $Y[k]$ need to be at least of length $N = L_y$.

Save the resulting audio as 'output.wav'.

f) Analytically determine if the echo system is causal and stable.

Deliverables:

1. One **uncompressed pdf** project report containing:
 - a. Explanation of your work.
 - b. All the required results and answers to questions.
 - c. All the required figures. Label your figures properly.
 - d. All the codes, included at the end.
2. All the codes files and required audio files.

Instructions:

- You can work in teams of up to 2 members per team.
- Any copied results or codes will result in zero grade for both teams.
- Code in the report should be supplied as text, not as screenshots.

Due date: May 9, 2025, at 11:59 PM.